

Quantum Key Distribution in the Post-Quantum Transition: What Security Is Physical, and What Remains Computational

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Quantum computation does not make cryptography obsolete; it changes which security assumptions remain defensible. Shor’s algorithm threatens the factorisation and discrete-logarithm problems underlying widely deployed public-key systems, whereas symmetric cryptography is affected more moderately by quantum search. Quantum key distribution (QKD) addresses a different problem. Rather than replacing encryption, it uses quantum states to establish shared key material while placing a measurable bound on information available to an eavesdropper. This distinction is frequently obscured by descriptions of QKD as “unhackable encryption.”

This review develops the physical argument from the BB84 protocol. We derive the 25 % quantum bit error rate produced by an ideal intercept–resend attack, state a composable secrecy criterion, and recover the asymptotic single-photon BB84 key fraction $r \geq 1 - 2h_2(Q)$. We then examine the assumptions lost when ideal qubits are replaced by attenuated laser pulses, optical fibre, and real detectors, including photon-number-splitting attacks, decoy states, detector side channels, and channel loss. Finally, QKD is placed beside the NIST post-quantum standards. The resulting picture is less dramatic than the phrase “quantum-secure communication” suggests, but more useful: post-quantum cryptography and QKD protect against different failure modes and should be evaluated as complementary technologies rather than interchangeable replacements.

I. INTRODUCTION

The security question raised by quantum computing is often stated too broadly. A sufficiently capable fault-tolerant quantum computer would not simply “break encryption.” It would invalidate particular computational assumptions on which important cryptographic protocols depend. RSA relies on the practical difficulty of integer factorisation [2], while Diffie–Hellman and elliptic-curve systems rely on discrete-logarithm problems [1]. Shor showed that both problem classes admit polynomial-time quantum algorithms [3]. The consequence is serious because public-key cryptography is deeply embedded in authentication, software signing, key establishment, and the protocols that secure network traffic.

The threat is also temporally asymmetric. Data encrypted today can be copied without being decrypted. If that information must remain confidential for longer than the time required to build a cryptographically relevant quantum computer, an adversary can store the ciphertext and attack it later. Migration therefore has to occur before such a machine exists, not after it has been demonstrated.

Quantum key distribution addresses this problem from a different direction. The best-known protocol, BB84, encodes classical bits into non-orthogonal quantum states [5]. An adversary attempting to acquire information about those states cannot in general determine which measurement basis was used without introducing statistical changes to the signals. Alice and Bob can estimate those changes and, provided that the measured disturbance is small enough, perform error correction and privacy amplification to obtain a shorter shared key about which the adversary has negligible information [6, 9].

That statement is more precise, and less sweeping, than the familiar claim that quantum cryptography is “unhackable.” QKD does not encrypt the application data, does not authenticate the communicating parties by itself, does not prevent denial of service, and does not remove the need to trust the physical implementation. The distinction between a security proof and the apparatus satisfying the assumptions of that proof is central to practical QKD [8, 13].

Meanwhile, the classical cryptographic response has advanced substantially. In 2024, the National Institute of Standards and Technology (NIST) finalised ML-KEM for key establishment and ML-DSA and SLH-DSA for digital signatures [16–18]. These algorithms are designed for deployment on conventional computing and communications infrastructure. Consequently, the relevant engineering question is no longer whether “classical” or “quantum” cryptography wins. It is which security mechanism is appropriate for a given threat model.

This paper develops that comparison from the underlying physics. We first identify which cryptographic assumptions are changed by quantum algorithms. We then derive the disturbance introduced by eavesdropping in BB84 and connect the measured quantum bit error rate to an ideal asymptotic key rate. The analysis is extended to weak coherent pulses and lossy optical channels before QKD is compared with post-quantum cryptography (PQC) as an engineered security system. No new experimental data are claimed; the objective is to make the boundary between a physical security statement and an implementation claim explicit.

II. THEORETICAL FRAMEWORK

A. Quantum computation changes the threat model

The distinction between classical and quantum cryptanalysis is most clearly seen in their scaling. Let N denote the integer to be factored and let $n = \log_2 N$ denote its binary length. The general number field sieve, the best-known classical algorithm for factoring sufficiently large generic integers, has sub-exponential asymptotic complexity. Suppressing constants, its running time can be written in the form

$$T_{\text{GNFS}}(n) = \exp\left[O\left(n^{1/3}(\ln n)^{2/3}\right)\right], \quad (1)$$

where T_{GNFS} is a measure of the number of computational operations. The important feature of [equation \(1\)](#) is not a particular calendar time but the scaling with the key size.

Shor's algorithm changes the scaling class. For ideal fault-tolerant quantum computation, integer factorisation and discrete logarithms can be solved using a number of elementary quantum operations that is polynomial in the input length,

$$T_{\text{Shor}}(n) = \text{poly}(n). \quad (2)$$

The transition from [equation \(1\)](#) to [equation \(2\)](#) is why simply increasing an RSA modulus does not provide the same type of protection against a future quantum computer.

Symmetric cryptography is affected differently. For an ideal k -bit key, exhaustive classical search requires $O(2^k)$ evaluations. Grover's quantum search algorithm reduces the query complexity to

$$T_{\text{Grover}}(k) = O\left(2^{k/2}\right), \quad (3)$$

where k is the number of key bits [\[4\]](#). This is a quadratic speed-up rather than the change of complexity class produced by Shor's algorithm. A 256-bit symmetric key therefore retains a large quantum search margin, although a complete security analysis must also account for the cost of implementing the oracle and error-corrected quantum computation.

The first conclusion is therefore narrow but important: the immediate cryptographic problem created by quantum computation is concentrated in public-key mechanisms based on factorisation and discrete logarithms, not in all classical cryptography.

B. The BB84 state space

QKD approaches key establishment without assuming that a particular mathematical problem is computationally difficult. In BB84, Alice encodes each classical bit into one of two mutually unbiased bases of a two-dimensional Hilbert space. We denote the computational basis by Z ,

$$\mathcal{B}_Z = \{|0\rangle, |1\rangle\}, \quad (4)$$

and the diagonal basis by X ,

$$\mathcal{B}_X = \{|+\rangle, |-\rangle\}, \quad |\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}. \quad (5)$$

Here $|0\rangle$ and $|1\rangle$ are orthogonal qubit states and $|+\rangle, |-\rangle$ are their equal superpositions.

The two bases are mutually unbiased because the squared overlap between any state in one basis and any state in the other is

$$|\langle z|x\rangle|^2 = \frac{1}{2}, \quad z \in \{0, 1\}, \quad x \in \{+, -\}. \quad (6)$$

[Equation \(6\)](#) means that a measurement in the wrong basis gives no information about the encoded bit. This complementarity, together with the impossibility of perfectly cloning an unknown quantum state, is the relevant physical structure behind BB84. Describing the protocol only as an example of the ‘‘observer effect’’ loses the feature that can actually be quantified.

TABLE I. BB84 encoding convention used in this paper. Alice chooses a classical bit and a basis independently. Bob later chooses a measurement basis independently. Only events for which Alice and Bob used the same basis contribute to the sifted key.

Classical bit	Basis	Transmitted state	Polarisation example
0	Z	$ 0\rangle$	horizontal
1	Z	$ 1\rangle$	vertical
0	X	$ +\rangle$	$+45^\circ$
1	X	$ -\rangle$	-45°

C. Security is an operational statement

Detecting disturbance is not itself the definition of a secure key. Modern QKD security is formulated operationally so that the resulting key can be safely used inside a larger cryptographic protocol [9]. Let K_A and K_B be the keys held by Alice and Bob after post-processing. Correctness requires that the probability of their keys disagreeing is bounded by a small parameter ε_{cor} ,

$$\Pr[K_A \neq K_B] \leq \varepsilon_{\text{cor}}. \quad (7)$$

A useful secrecy condition compares the actual joint state of the key and the eavesdropper with an ideal state in which the key is uniform and independent. It can be written as

$$\frac{1}{2} \|\rho_{K_A E} - \tau_{K_A} \otimes \rho_E\|_1 \leq \varepsilon_{\text{sec}}, \quad (8)$$

where $\rho_{K_A E}$ is the actual classical–quantum state, τ_{K_A} is a perfectly uniform key state, ρ_E is Eve’s reduced state, and $\|\cdot\|_1$ denotes the trace norm. A protocol satisfying equations (7) and (8) is secure up to a failure parameter of order $\varepsilon_{\text{cor}} + \varepsilon_{\text{sec}}$.

This definition clarifies what the physics is used to prove. QKD does not promise that Eve never interacts with the channel. It promises that, conditioned on the protocol accepting, Alice and Bob can bound how far their final key is from an ideal secret key.

III. MAIN DERIVATION AND METHODS

A. From state preparation to a sifted key

For every transmission round, Alice chooses a uniformly random bit and a uniformly random basis, prepares the corresponding state in table I, and sends it through the quantum channel. Bob independently chooses either the Z or X basis and records his measurement outcome. After many rounds, Alice and Bob announce their basis choices over an authenticated classical channel but do not disclose the corresponding key bits. Events measured in different bases are discarded.

For unbiased basis choices, Alice and Bob agree on a basis with probability

$$P_{\text{sift}} = P(ZZ) + P(XX) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}. \quad (9)$$

Thus, in ideal BB84 approximately half of all detected signals survive basis sifting. Modern variants can bias the basis probabilities so that the asymptotic sifting efficiency approaches unity while retaining enough measurements in the complementary basis for parameter estimation.

B. Intercept–resend attack and its error signature

The simplest attack illustrates why non-orthogonal states matter. Suppose Eve intercepts every signal, chooses Z or X with equal probability, measures the photon, prepares a new state corresponding to her result, and sends that state to Bob. We calculate the error probability only on sifted events, for which Alice and Bob happened to choose the same basis.

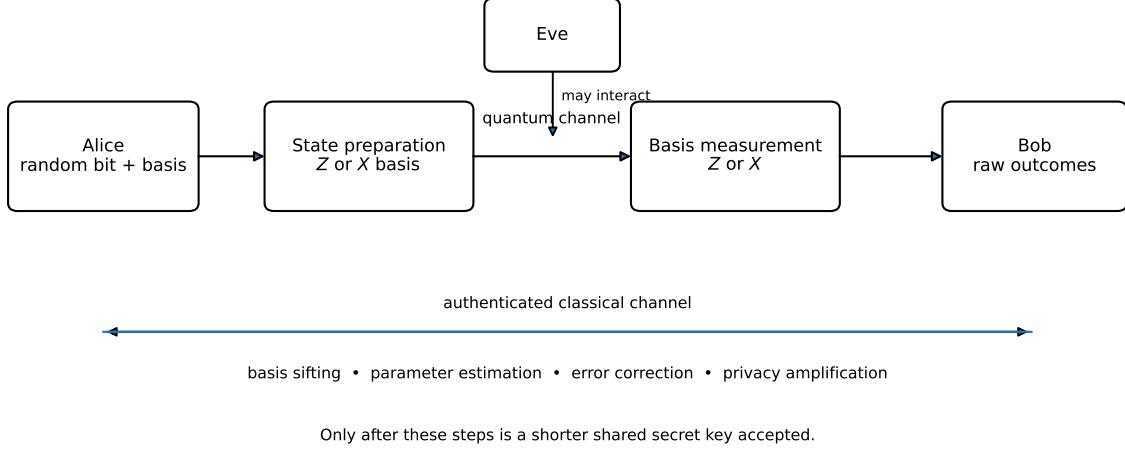


FIG. 1. Operational structure of BB84. Alice prepares a random state in either the Z or X basis and Bob measures in a randomly selected basis. The quantum channel is assumed to be fully accessible to an adversary, whereas the classical channel must be authenticated. Basis sifting, parameter estimation, error correction, and privacy amplification convert the raw detection record into a shorter secret key.

Eve chooses the wrong basis with probability $1/2$. Conditional on that event, [equation \(6\)](#) implies that her outcome is random with respect to Alice's bit. The state she resends is therefore also measured incorrectly by Bob with probability $1/2$. The resulting quantum bit error rate (QBER) is

$$Q_{\text{IR}} = P(\text{Eve wrong basis})P(\text{Bob error} \mid \text{Eve wrong basis}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}. \quad (10)$$

An ideal full intercept-resend attack therefore introduces a 25 % QBER into the sifted key. The complete conditional probability bookkeeping is given in [section B](#).

Equation [equation \(10\)](#) should not be interpreted as a universal 25 % attack-detection threshold. Real adversaries can perform more general interactions, and real systems contain background counts, misalignment, finite sample sizes, and device imperfections. Its value is pedagogical: acquiring information about randomly chosen non-orthogonal states has an experimentally observable statistical cost.

C. From measured errors to secret-key fraction

A useful ideal benchmark follows from the asymptotic security proof of BB84 [\[6\]](#). We first define the binary entropy function,

$$h_2(x) = -x \log_2 x - (1-x) \log_2 (1-x), \quad (11)$$

where x is a probability between zero and one. The function quantifies the binary uncertainty associated with an error rate x .

For an ideal single-photon implementation, infinitely long keys, symmetric bit and phase error rates, and one-way classical post-processing, the secret fraction per sifted bit satisfies

$$r \geq 1 - 2h_2(Q), \quad (12)$$

where Q is the measured QBER. One term $h_2(Q)$ accounts for information that must be revealed during error correction, while the second reflects the privacy amplification required to suppress Eve's information.

The right-hand side of [equation \(12\)](#) reaches zero at approximately

$$Q_{\text{th}} \simeq 0.1100. \quad (13)$$

Thus the familiar approximately 11 % value is a property of this specific asymptotic security model, not a universal threshold for every QKD protocol. Finite-key corrections, detector statistics, biased bases, and different post-processing schemes change the practical acceptance criterion.

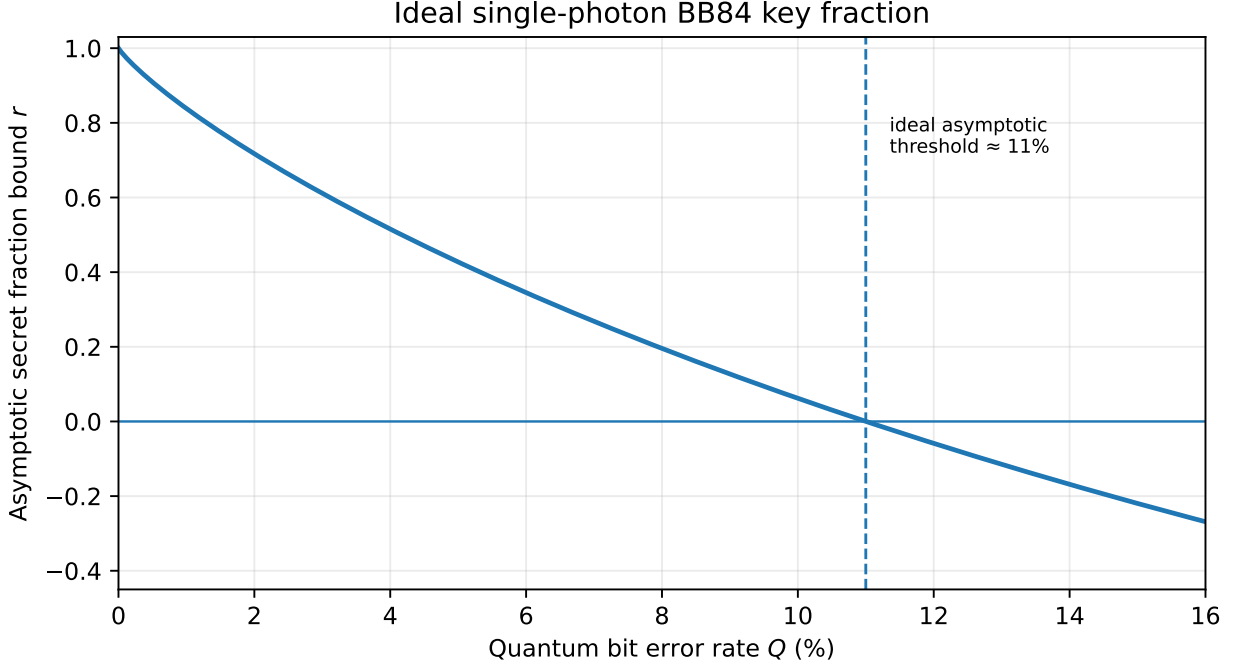


FIG. 2. Asymptotic single-photon BB84 secret-fraction bound from [equation \(12\)](#). The bound falls to zero at a QBER of approximately 11 % under the assumptions stated in the text. The figure is a theoretical benchmark, not a universal operating threshold for all QKD protocols.

D. Weak coherent pulses and decoy states

Ideal single-photon sources are difficult to operate at the rates and wavelengths required for telecommunications. Practical BB84 systems therefore often attenuate coherent laser pulses so that the mean photon number μ is below unity. The photon-number distribution of a phase-randomised coherent pulse is Poissonian,

$$P_\mu(n) = e^{-\mu} \frac{\mu^n}{n!}, \quad (14)$$

where $P_\mu(n)$ is the probability that a pulse of mean intensity μ contains exactly n photons.

Equation [equation \(14\)](#) exposes an implementation problem absent from the ideal qubit model. Some pulses contain more than one photon. In a photon-number-splitting attack, Eve can in principle remove one photon from a multiphoton pulse and delay its measurement until the basis has been announced, while forwarding the remaining photons to Bob. Channel loss can conceal the removed signals.

Decoy-state QKD addresses this vulnerability by randomly varying the pulse intensity [\[10, 11\]](#). Because Eve does not know whether a particular photon-number component originated from a signal or decoy intensity, Alice and Bob can use the observed detection statistics to bound the yield and error rate of the single-photon contribution.

A commonly used asymptotic lower bound on the decoy-state BB84 key rate can be written

$$R \geq q [-Q_\mu f_{\text{EC}} h_2(E_\mu) + Q_1 (1 - h_2(e_1))], \quad (15)$$

where R is the secret-key rate per emitted pulse, q is the basis-sifting factor, Q_μ is the total signal gain, E_μ is the measured signal QBER, $f_{\text{EC}} \geq 1$ quantifies the inefficiency of practical error correction, Q_1 is the gain of the single-photon component, and e_1 is its estimated error rate. The importance of [equation \(15\)](#) is conceptual: practical security is recovered by estimating which subset of the optical data still satisfies a provable single-photon security model.

IV. RESULTS

The preceding derivation produces three results relevant to the engineering of quantum-secure communications. First, information acquisition in BB84 is linked to a measurable disturbance rather than to an assumed computational

TABLE II. Ideal channel attenuation calculated from [equation \(16\)](#) for $\alpha = 0.2$ dB/km. The table includes fibre loss only; practical systems incur additional source, coupling, and detector losses.

Distance L (km)	Channel loss (dB)	Transmittance η_{ch}
50	10	10^{-1}
100	20	10^{-2}
200	40	10^{-4}
500	100	10^{-10}

cost. Second, a low QBER is useful only after it is translated into a secrecy bound under a specified device model. Third, the usable key rate is strongly constrained by the probability that a quantum signal survives the physical channel.

A. Optical loss sets the distance scale

For standard optical fibre, attenuation is conveniently expressed in decibels per unit length. Let α denote the attenuation coefficient and L the fibre length. The channel transmittance is

$$\eta_{\text{ch}}(L) = 10^{-\alpha L/10}. \quad (16)$$

Here η_{ch} is the fraction of optical power that survives the channel in the absence of other losses. Around the telecommunications C band, a representative value for conventional low-loss fibre is approximately $\alpha = 0.2$ dB/km.

The exponential consequence of [equation \(16\)](#) is shown in [table II](#). At 100 km, only about one photon in one hundred survives the fibre attenuation alone. At 500 km, the nominal transmittance is of order 1×10^{-10} before detector inefficiency, coupling loss, or background noise is included.

This estimate provides an immediate sanity check on long-distance claims. Secure fibre QKD has nevertheless been demonstrated over 421 km using ultralow-loss fibre and superconducting single-photon detectors [15]. The extremely small rates at such distances are not a failure of the security principle; they are a consequence of transmitting non-amplifiable quantum states through a lossy channel.

A satellite changes the geometry. Most of the optical path then lies through near-vacuum rather than fibre, so the loss is dominated by diffraction, pointing, atmospheric transmission, and detector efficiency rather than by an exponential fibre attenuation coefficient. The Micius experiment demonstrated decoy-state satellite-to-ground QKD over distances up to approximately 1200 km, with secret-key generation at kilohertz scale during suitable links [14]. The result demonstrates that continental distance is physically possible, but it does not remove the need for trusted ground infrastructure, authentication, or a suitable satellite access window.

B. The device model is part of the security proof

The ideal BB84 derivation treats state preparation and measurement as known quantum operations. Real hardware can depart from that model in ways available to an adversary. Timing-dependent detector efficiencies, bright-light control, imperfect phase modulation, information leaked from the source, or incorrectly characterised intensity fluctuations can create side channels that are not represented by the ideal qubit description [8, 13].

This observation does not invalidate the theoretical security of QKD. It changes the experimental task. A deployed device must be shown to satisfy the assumptions entering the proof, or the protocol must be modified so that security depends on fewer assumptions.

Measurement-device-independent QKD (MDI-QKD) is an important example [12]. Alice and Bob each prepare states and send them to an untrusted central measurement station. Security is constructed so that attacks on the measurement devices are included in the adversarial model. The price is greater experimental complexity, stringent interference requirements, and typically lower rates than short-range point-to-point BB84.

Device-independent QKD pushes this idea further by attempting to infer security from observed nonlocal correlations rather than a detailed internal model of the devices. Its assumptions are conceptually weaker but experimentally much more demanding. “Device independent” therefore does not mean “assumption free.” Isolation of laboratories, random-number generation, causal structure, and the validity of quantum mechanics remain part of the security statement.

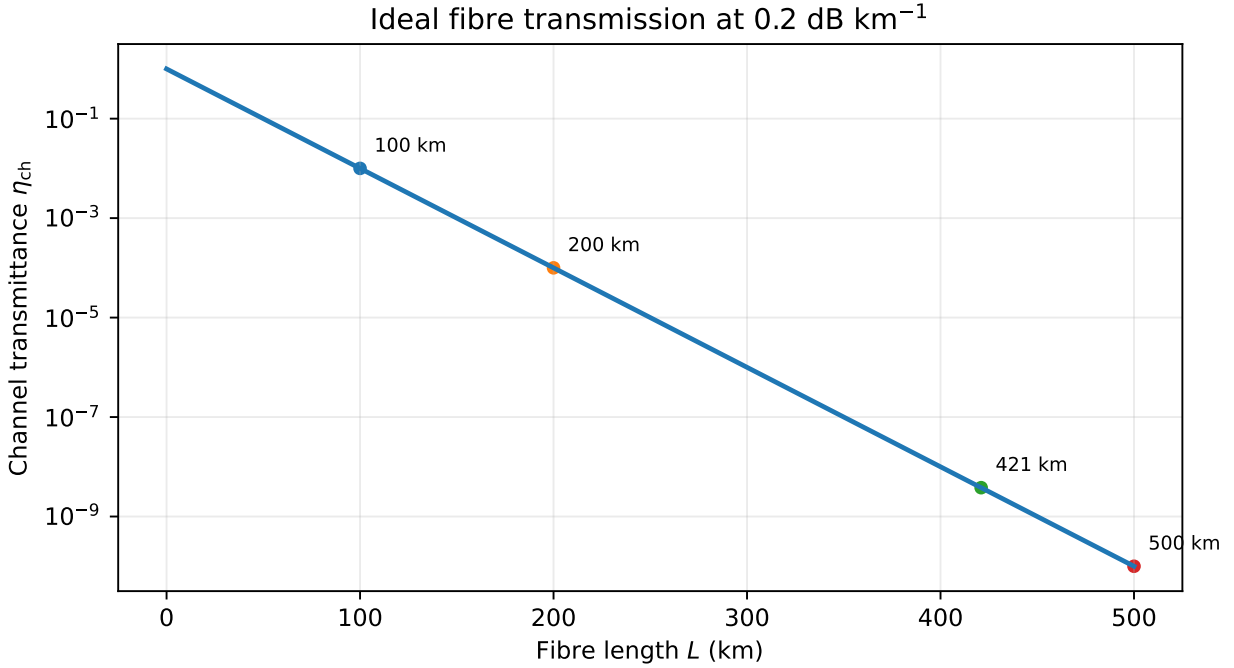


FIG. 3. Calculated transmission through optical fibre with attenuation $\alpha = 0.2$ dB/km, using [equation \(16\)](#). The vertical axis should be logarithmic. The exponential decrease in detected signal is the central engineering constraint on direct long-distance fibre QKD and motivates improved detectors, decoy-state protocols, alternative geometries, and future quantum repeater architectures.

C. The post-quantum comparison

The corresponding classical migration path is now concrete. NIST FIPS 203 standardises ML-KEM for key establishment, while FIPS 204 and FIPS 205 standardise ML-DSA and SLH-DSA for signatures [16–18]. NIST selected the code-based HQC algorithm in 2025 as an additional key-establishment candidate with a different mathematical foundation from ML-KEM; its final standardisation is expected later in the process [19].

The relevant contrast is summarised in [table III](#). The two approaches do not protect information in the same way and should not be presented as equivalent technologies with different branding.

V. DISCUSSION

A. What the laws of physics actually provide

The strongest defensible statement about QKD is not that it is impossible to hack. It is that quantum theory allows the secrecy of a generated key to be bounded without imposing a computational limit on the eavesdropper, provided that the protocol and physical devices satisfy the assumptions used in the security proof [9]. Eve may possess unlimited classical computing power, quantum memory, and the ability to control the quantum channel. Security is inferred from the observed statistics and the model.

This distinction matters because theoretical adversarial power and implementation trust move in opposite directions. A proof can grant Eve complete control of the transmission channel while still requiring Alice’s source to have a specified photon-number distribution or Bob’s detector to behave within a characterised model. If those assumptions fail, the mathematical proof may remain correct while no longer describing the experiment.

The practical-security literature has therefore moved from the slogan “security guaranteed by physics” toward explicit device characterisation, decoy-state analysis, MDI-QKD, finite-key statistics, and composable security definitions [8, 9, 13]. This is a sign of maturity rather than weakness: the assumptions have become objects that can be stated, measured, attacked, and improved.

TABLE III. Comparison of QKD and post-quantum cryptography at the system level. “Physical security” for QKD means security derived from quantum-mechanical constraints under stated assumptions about devices and laboratories; it does not imply that an implementation is automatically secure.

Criterion	Quantum key distribution	Post-quantum cryptography
Primary task	Generation or distribution of shared key material	Key establishment, signatures, authentication, and related cryptographic tasks
Security foundation	Quantum-mechanical constraints plus an explicit implementation model	Computational hardness of problems for which no efficient classical or quantum attack is known
Infrastructure	Quantum optical transmitters, detectors, and a suitable fibre or free-space channel	Conventional processors and existing digital networks
Authentication	Requires an authenticated classical channel; QKD does not establish initial identity by itself	Can provide digital signatures and authenticated key establishment
Upgrade path	Often hardware dependent	Primarily software, protocol, firmware, and certificate migration
Distance constraint	Strongly affected by optical loss; trusted nodes, satellites, MDI schemes, or future repeaters may extend reach	No quantum-channel distance penalty beyond the ordinary network
Failure mode	Mismatch between proof assumptions and physical devices; channel disruption; trusted-node compromise	Unexpected cryptanalytic breakthrough, implementation bug, weak randomness, or protocol misuse

B. QKD is not a replacement for authentication

BB84 requires a classical channel on which Alice and Bob can verify the origin and integrity of messages. Without authentication, an adversary can impersonate Bob to Alice and Alice to Bob, creating two independent QKD sessions. The quantum channel alone does not distinguish this man-in-the-middle attack from legitimate communication.

The authentication requirement can be satisfied using an initially shared secret and information-theoretic message authentication, or by a digital signature scheme. During the post-quantum transition, the latter can itself be implemented using quantum-resistant signatures. QKD can then refresh symmetric key material after identity has been established.

This dependency is one reason that QKD should be described as a key-management technology rather than a complete cryptographic stack. Confidentiality, integrity, identity, availability, endpoint security, and software assurance remain separate system requirements.

C. Availability is different from secrecy

The same sensitivity that makes interference detectable also makes QKD easy to disrupt. An adversary who injects noise, blocks photons, misaligns a free-space link, or simply cuts the fibre can force the estimated QBER above the acceptance threshold. Alice and Bob can then refuse to generate a key, which protects secrecy but not service availability.

No cryptographic protocol can prevent a severed physical link from being severed. The point is nevertheless important because “detectable eavesdropping” is sometimes interpreted as “attack prevention.” QKD converts some confidentiality attacks into detectable protocol failure; it does not guarantee continued communication.

D. Where QKD is technically justified

For the majority of internet infrastructure, PQC has a decisive deployment advantage. It uses existing processors, switches, certificate systems, and network links, and it can address both key establishment and digital signatures. NIST explicitly recommends migration toward its standardised PQC algorithms, while the NSA currently does not recommend QKD for protecting U.S. National Security Systems unless several implementation and infrastructure limitations are overcome [20]. The latter is a system-engineering judgement rather than a statement that the quantum physics is incorrect.

QKD becomes more interesting when the security requirement and network topology justify dedicated physical infrastructure. Examples include high-value fixed links, data-centre interconnects, or specialised government and

scientific networks in which optical hardware is already controlled and the cost of key generation is small compared with the value or lifetime of the protected information. In such cases, the different failure basis of QKD may be valuable even when PQC is also deployed.

A hybrid design can deliberately avoid making either mechanism a single point of cryptographic failure. One conceptual construction combines independently established secrets before deriving the application key,

$$K_{\text{session}} = \text{KDF}(K_{\text{PQC}} \parallel K_{\text{QKD}}), \quad (17)$$

where K_{PQC} is key material obtained from a post-quantum key-encapsulation mechanism, K_{QKD} is QKD-derived material, $\parallel$ denotes concatenation, and KDF is a suitable key-derivation function. Equation (17) is schematic rather than a complete protocol specification; the security of a real hybrid construction depends on the combiner, authentication, failure handling, and independence of the relevant assumptions.

The architectural point is more important than the particular formula. PQC hedges against the expense and topology restrictions of quantum links, while QKD can introduce a security mechanism whose core secrecy argument is not based on the computational hardness assumptions used by the PQC component.

VI. CONCLUSION

Quantum computing changes cryptography by changing which assumptions remain safe, not by making secrecy impossible. Shor’s algorithm directly challenges the factorisation and discrete-logarithm assumptions used by major public-key systems, whereas symmetric cryptography experiences a less destructive quadratic search speed-up. The practical response has therefore already begun: post-quantum algorithms are being standardised and deployed before a cryptographically relevant quantum computer exists.

Quantum key distribution offers a different security primitive. In BB84, information is encoded into non-orthogonal states so that eavesdropping can alter measurable channel statistics. For a full intercept–resend attack, the ideal disturbance is a 25 % QBER. In the asymptotic single-photon model, the observed QBER enters directly into a provable secret-key fraction. These statements give the phrase “security from quantum mechanics” a quantitative meaning.

The transition from that ideal model to a useful system is the difficult part. Weak coherent pulses introduce multiphoton components, optical fibre introduces exponential loss, detectors introduce side channels, and QKD still requires authenticated classical communication. Decoy states and measurement-device-independent protocols demonstrate how security theory can be redesigned around these imperfections rather than ignoring them.

The most credible role for QKD is therefore not as a universal replacement for classical cryptography. PQC provides the scalable migration path for general digital infrastructure. QKD provides a specialised physical mechanism for establishing keys under a different class of assumptions. Where the threat model and infrastructure justify it, combining the two may be more valuable than treating them as competitors.

Future progress will depend less on stronger claims of “unhackability” and more on reducing the gap between mathematical security models and physical devices: tighter finite-key proofs, better characterised sources and detectors, measurement-device-independent architectures, improved long-distance links, and ultimately practical quantum repeaters. The central question is no longer whether quantum mechanics can contribute to cryptography. It is how precisely we can state, engineer, and verify the security that it contributes.

Appendix A: Notation

Appendix B: Conditional derivation of the intercept–resend QBER

We derive equation (10) explicitly to show where the factor of 1/4 originates. Condition on a sifted event, so Alice and Bob used the same basis. Eve has two possibilities.

If Eve happened to choose Alice’s basis, which occurs with probability 1/2, she measures the correct state with unit probability in the ideal model. She resends the same state and Bob obtains Alice’s original bit. Thus

$$P(\text{error} \mid E = A) = 0, \quad (B1)$$

where $E = A$ denotes Eve choosing Alice’s basis.

If Eve chose the complementary basis, which also occurs with probability 1/2, the mutual-unbiasedness relation in equation (6) makes her outcome uniformly random. Consider, for example, Alice transmitting $|0\rangle$ while Eve measures in the X basis. The outcome probabilities are

$$P(+ \mid 0) = |\langle +|0\rangle|^2 = \frac{1}{2}, \quad P(- \mid 0) = |\langle -|0\rangle|^2 = \frac{1}{2}. \quad (B2)$$

TABLE IV. Principal symbols used throughout the paper.

Symbol	Definition
N	Integer to be factored in the classical/quantum complexity comparison
n	Binary input length, $n = \log_2 N$
k	Number of bits in a symmetric cryptographic key
$\mathcal{B}_Z$	Computational basis $\{ 0\rangle, 1\rangle\}$
$\mathcal{B}_X$	Diagonal basis $\{ +\rangle, -\rangle\}$
Q	Quantum bit error rate of the sifted key
r	Secret fraction per sifted bit in the ideal asymptotic model
$h_2(x)$	Binary entropy function
μ	Mean photon number of a phase-randomised coherent pulse
Q_μ	Detection gain for signal intensity μ
E_μ	Measured QBER for signal intensity μ
Q_1	Single-photon gain estimated using decoy statistics
e_1	Error rate of the estimated single-photon component
α	Optical-fibre attenuation coefficient
L	Optical channel length
η_{ch}	Fibre channel transmittance
ε_{cor}	Correctness failure probability
ε_{sec}	Secrecy failure parameter

Eve therefore sends either $|+\rangle$ or $|-\rangle$ to Bob with equal probability.

Bob measures that resent state in Alice's original Z basis. For either $|+\rangle$ or $|-\rangle$, the probability of recovering Alice's bit is $1/2$, so

$$P(\text{error} \mid E \neq A) = \frac{1}{2}. \quad (\text{B3})$$

The law of total probability then gives

$$\begin{aligned} Q_{\text{IR}} &= P(E = A)P(\text{error} \mid E = A) + P(E \neq A)P(\text{error} \mid E \neq A) \\ &= \frac{1}{2}(0) + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{1}{4}. \end{aligned} \quad (\text{B4})$$

The same calculation holds for all four BB84 signal states by symmetry.

Appendix C: Sanity check on the ideal BB84 key threshold

The threshold quoted in [equation \(13\)](#) follows by setting the asymptotic secret fraction in [equation \(12\)](#) to zero,

$$1 - 2h_2(Q_{\text{th}}) = 0. \quad (\text{C1})$$

Hence the threshold satisfies

$$h_2(Q_{\text{th}}) = \frac{1}{2}. \quad (\text{C2})$$

Numerically solving [equation \(C2\)](#) gives $Q_{\text{th}} \simeq 0.1100$.

This result passes two immediate limiting checks. At $Q = 0$, $h_2(0) = 0$ and [equation \(12\)](#) gives $r = 1$, as expected for an ideal error-free sifted key. At $Q = 1/2$, $h_2(1/2) = 1$ and the lower bound becomes negative, signalling that no secret key can be certified by this expression. A practical protocol replaces a negative bound with zero and aborts.

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